DAY 16 — AI FOR DECISION MAKING

Learning outcome: participants can use AI to support data-driven decisions, explain evidence, and verify recommendations.

(□ ChatGPT □ Gemini □ Claude □ Copilot □ Perplexity)

Hour 1: What is AI-assisted decision making?

Theory:

- A decision is a choice between two or more possible actions.
- AI-assisted decision-making means using AI to organize information, identify patterns, compare options, and draft recommendations.
- AI is a support tool—not the final decision-maker.
- Humans set goals, check evidence, consider values, and accept responsibility.

Decision cycle:

1. Define the question
2. Gather information
3. Compare options
4. Choose and act
5. Review the result

Key terms:

- Option: a possible action
- Criterion: a standard used to compare options
- Evidence: facts or data supporting a conclusion
- Assumption: something accepted as true for planning
- Uncertainty: something we do not know for sure

Example:

Question: “Which day should we run a student workshop?”

Options: Friday or Saturday

Criteria: attendance, room cost, trainer availability

Important rule:

Never accept an AI answer without checking the facts and assumptions.

Practice :

Choose one simple decision:

- Buy a laptop A or laptop B
- Select Friday or Saturday for an event
- Choose online or in-person training

Complete this template:

- My decision is: _____
- My two options are: _____ and _____
- My three criteria are: _____, _____, _____
- One constraint is: _____
- One thing I need to verify is: _____

Easy AI prompt:

"I need to decide between [Option A] and [Option B]. My criteria are [list criteria]. Ask me three questions before you recommend an option."

Check your AI result:

- Did it understand the two options?
- Did it use your criteria?
- Did it ask useful questions?
- Did it invent any information?

Pair discussion :

Explain your decision and one fact that must be checked.

Hour 2: Data-driven decision making

Theory:

- Data-driven decisions use measured information instead of only opinion or guesswork.
- Data can be numbers (sales, attendance) or categories/text (department, feedback).
- A KPI is a Key Performance Indicator: a number that shows progress toward a goal.

Four types of analysis:

- Descriptive: What happened?
- Diagnostic: Why did it happen?
- Predictive: What may happen next?
- Prescriptive: What should we do?

Use AI to explain, organize, or compare data—not to invent missing values.

Data-quality basics:

- Complete: important values are not missing
- Accurate: values match the real situation
- Consistent: names and formats are the same
- Timely: data is recent enough for the decision

Useful formulas:

- Conversion rate = enrollments / leads × 100
- Profit = revenue − cost
- Average = total / number of items

Example:

100 leads and 20 enrollments = 20% conversion rate.

AI can explain what the rate means, but you should calculate or verify it.

Hour 2: Easy practice — calculate and interpret

Practice dataset:

- Leads: 100
- Enrollments: 20
- Course fee per enrollment: 2,000 BDT
- Marketing cost: 10,000 BDT

Tasks:

1. Calculate conversion rate.
2. Calculate total revenue.
3. Calculate profit (ignore other costs).
4. Write one sentence: Is the result good or bad? What extra information do you need?

Expected calculations:

- Conversion rate = 20 / 100 × 100 = 20%
- Revenue = 20 × 2,000 = 40,000 BDT
- Profit = 40,000 − 10,000 = 30,000 BDT

Easy AI prompt:

"Using these figures—100 leads, 20 enrollments, course fee 2,000 BDT, marketing cost 10,000 BDT—show the formulas for conversion rate, revenue, and profit. Do not add assumptions. Then give a simple interpretation."

Verification task:

Check whether the AI used the exact figures and formulas.

Theory:

- Business intelligence (BI) turns operational data into useful reports and dashboards.
- A dashboard gives a quick view of important performance measures.
- Good dashboards answer a decision question, not merely display many charts.

Dashboard elements:

- KPI cards: key numbers such as total enrollment
- Trend: change over time
- Target: expected result
- Comparison: actual vs. previous month or plan
- Filter: view by course, month, location, or team

How to read a dashboard:

1. Read the title and time period.
2. Identify the KPI and its definition.
3. Compare actual result with target or previous result.
4. Look for the largest change.
5. Ask what action is needed.

Example:

Enrollment target = 50; actual enrollment = 40.

Gap = 10 students below target.

Next question: Which course or channel caused the gap?

AI role:

AI can summarize trends and suggest questions; confirm every statement using the dashboard data.

Hour 3: Easy practice — read a mini dashboard

Mini dashboard data:

- January enrollment: 40; target: 50
- February enrollment: 55; target: 50
- January attendance: 82%
- February attendance: 75%

Answer these questions:

1. Which month met the enrollment target?
2. How much did enrollment change from January to February?
3. What happened to attendance?
4. Write one question a manager should investigate.

Possible answers:

- February met the target.
- Enrollment increased by 15 students.
- Attendance fell from 82% to 75% (a 7 percentage-point drop).
- Investigation question: Why did attendance fall while enrollment rose?

Easy AI prompt:

“Summarize this dashboard in three short bullets. Separate facts from questions to investigate. Do not claim a cause unless the data proves it.”

Class discussion:

Why is “attendance fell because enrollment rose” not a proven conclusion?

Theory: financial decisions

- Revenue: money earned from sales or services.
- Cost: money spent to operate or deliver a service.
- Profit: revenue minus cost.
- Break-even: the point where revenue equals cost.
- Scenario: a possible future situation, such as best case or worst case.

Theory: risk decisions

- A risk is an uncertain event that can affect the result.
- Likelihood: how likely it is to happen.
- Impact: how serious the effect would be.
- Mitigation: an action that reduces likelihood or impact.

Simple risk matrix:

- Low likelihood + low impact = monitor
- High likelihood + low impact = manage routinely
- Low likelihood + high impact = prepare a contingency
- High likelihood + high impact = urgent action

Example:

Risk: trainer may be unavailable.

Likelihood: medium; impact: high.

Mitigation: identify a backup trainer before launching.

AI safety note:

Do not paste confidential financial or personal data into tools unless approved by your organization.

Hour 4: Easy practice — compare two options

Option A: Online workshop

- Expected revenue: 30,000 BDT
- Expected cost: 8,000 BDT
- Main risk: internet disruption

Option B: In-person workshop

- Expected revenue: 40,000 BDT
- Expected cost: 25,000 BDT
- Main risk: low attendance

Tasks:

1. Calculate profit for each option.
2. Select one option.
3. Give one reason and one risk-control action.

Expected calculations:

- Option A profit = 22,000 BDT
- Option B profit = 15,000 BDT

Example response:

“I choose Option A because expected profit is higher. To reduce internet risk, test the platform and prepare a recorded backup lesson.”

Easy AI prompt:

“Create a simple comparison table for Option A and Option B using revenue, cost, profit, main risk, and mitigation. Then give a cautious recommendation in two sentences.”

Theory:

• A decision report converts analysis into a short, actionable recommendation.

• The report should help a manager decide quickly and understand the reason.

Basic report structure:

1. Decision needed
2. Recommendation
3. Evidence
4. Risks and assumptions
5. Next action, owner, and date

Good recommendation language:

• “Based on the available data, recommend…”

• “Confidence is moderate because…”

• “This recommendation may change if…”

Facts vs. assumptions:

• Fact: “February enrollment was 55.”

• Assumption: “Enrollment will stay above 50 next month.”

• Recommendation: “Run another campaign next month.”

Quality checklist:

• Is the decision question clear?

• Are calculations correct?

• Are sources and dates stated?

• Are risks and uncertainty included?

• Is there a named action and deadline?

AI can draft the report; people must verify the figures and approve the action.

Hour 5: Easy practice — complete a decision report

Use this situation:

“February enrollment was 55, above the target of 50. Attendance was 75%, below January’s 82%. The center can either spend more on advertising or improve attendance support.”

Complete the five lines:

- Decision needed: _____
- Recommendation: _____
- Evidence: _____
- Risk/assumption: _____
- Next action and owner: _____

Example answer:

- Decision needed: choose the priority for March.
- Recommendation: improve attendance support first.
- Evidence: enrollment is above target, but attendance fell by 7 points.
- Risk/assumption: the reason for low attendance is not yet known.
- Next action: Student-support officer surveys absent learners by Friday.

Easy AI prompt:

“Write a five-line decision report from this information. Label facts, assumptions, and recommendation separately. Use clear, simple language.”

Theory: strategic planning

- Strategy is a long-term direction for achieving an important goal.
- A plan turns the strategy into activities, owners, dates, and measures.
- A forecast estimates what may happen in the future based on available information.

Planning components:

- Goal: what success looks like
- Initiative: the action to take
- Milestone: an important checkpoint
- Leading indicator: early sign of progress
- Review: when to check results and change the plan

Forecasting principles:

- Begin with a simple baseline.
- State the assumptions clearly.
- Use scenarios: base, optimistic, and downside.
- Compare forecast with actual results later.

Example:

Past enrollment: 40 in January, 55 in February.

Baseline forecast: 55 for March (carry forward the latest month).

Optimistic scenario: 65 if a campaign succeeds.

Downside scenario: 45 if attendance concerns reduce referrals.

Hour 6: Easy practice — three-month forecast

Given data:

- January: 40 enrollments
- February: 50 enrollments
- March: 60 enrollments

Easy forecast task:

1. Base forecast for April: use March's result = _____
2. Optimistic forecast: add 10 enrollments = _____
3. Downside forecast: subtract 10 enrollments = _____
4. Write one assumption for each scenario.

Expected answer:

- Base: 60 enrollments
- Optimistic: 70 enrollments
- Downside: 50 enrollments

Easy AI prompt:

"Using January 40, February 50, and March 60 enrollments, create base, optimistic, and downside April forecasts. State one simple assumption for each. Do not present the forecast as certain."

Final reflection:

What number will you monitor in April to check whether your forecast is accurate?

Wrap-up: responsible AI decision-making

Before using an AI recommendation, check:

- Did I define the decision clearly?
- Did I provide real, relevant data?
- Did I check calculations and source information?
- Did I separate facts from assumptions?
- Did I consider risk and an alternative view?
- Did I identify an owner, action, and review date?

Key message:

AI helps people think faster and organize information. Good decisions still require human judgment, verification, and accountability.

Exit ticket

1. Write one simple decision you need to make this week.
2. List two criteria you will use.
3. Write one AI prompt you can use.
4. Name one fact you must verify before acting.

Example prompt:

“Help me compare these two options using my criteria. Ask questions first, identify missing information, and do not invent data.”

Source alignment: Expanded from Day 16 of the supplied AI course module: AI decision making, data-driven decisions, BI and analytics, financial/risk decisions, dashboards/reports, and forecasting.